



## Semester Two Examination, 2021

### Question/Answer booklet

# MATHEMATICS APPLICATIONS UNITS 1&2

## Section Two: Calculator-assumed

# SOLUTIONS

WA student number: In figures

|  |  |  |  |  |  |  |  |
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In words

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Your name

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### Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Number of additional  
answer booklets used  
(if applicable):

|  |
|--|
|  |
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### Materials required/recommended for this section

#### *To be provided by the supervisor*

This Question/Answer booklet

Formula sheet (retained from Section One)

#### *To be provided by the candidate*

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

### Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

## Structure of this paper

| Section                         | Number of questions available | Number of questions to be answered | Working time (minutes) | Marks available | Percentage of examination |
|---------------------------------|-------------------------------|------------------------------------|------------------------|-----------------|---------------------------|
| Section One: Calculator-free    | 8                             | 8                                  | 50                     | 52              | 35                        |
| Section Two: Calculator-assumed | 13                            | 13                                 | 100                    | 98              | 65                        |
| Total                           |                               |                                    |                        |                 | 100                       |

## Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (98 Marks)

This section has **thirteen** questions. Answer **all** questions. Write your answers in the spaces provided.

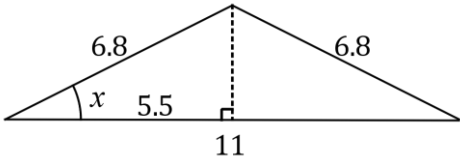
Working time: 100 minutes.

Question 9

(6 marks)

- (a) An isosceles triangle has two sides of length 6.8 cm and a base of length 11 cm. Sketch this triangle and hence determine the angle between the base and one of the sides.

(3 marks)

| Solution                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  $\cos x = \frac{5.5}{6.8} \quad \text{OR} \quad \cos x = \frac{6.8^2 + 11^2 - 6.8^2}{2(6.8)(11)}$ $x = 36.0^\circ$ |
| Specific behaviours                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>✓ sketch, indicates required angle</li> <li>✓ indicates expression for cosine of angle</li> <li>✓ correct angle</li> </ul>                                     |

- (b) Clearly show use of Heron's rule to determine the area of a triangle that has sides of length 16 cm, 23.1 cm and 28.1 cm.

(3 marks)

| Solution                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $s = \frac{14 + 17.1 + 22.1}{2} = 33.6$ $A = \sqrt{33.6(33.6 - 14)(33.6 - 17.1)(33.6 - 22.1)}$ $= \sqrt{33.6 \times 17.6 \times 10.5 \times 5.5}$ $= 184.8 \text{ cm}^2$ |
| Specific behaviours                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>✓ calculates semi-perimeter</li> <li>✓ substitutes correctly</li> <li>✓ calculates area</li> </ul>                                |

## Question 10

(5 marks)

- (a) The length of a hallway in an office is 8.4 metres. On a scale drawing of the office, the hallway has a length of 70 mm. Determine the scale used to make the drawing. Give your answer in the form 1:  $n$ , where  $n$  is a whole number. (2 marks)

| Solution                                                    |
|-------------------------------------------------------------|
| $70 \text{ mm} : 8.4 \text{ m}$<br>$70 : 8400$<br>$1 : 120$ |
| Specific behaviours                                         |
| ✓ ratio, using same units<br>✓ ratio in required form       |

- (b) A plan for an area outside the office is drawn to a scale of 1: 200, and on this plan, a rectangular courtyard is 4.8 cm wide and 11.4 cm long. If this courtyard is paved with square pavers measuring  $30 \times 30$  cm, determine how many pavers it contains. (3 marks)

| Solution                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $w = 4.8 \times 200 = 960 \text{ cm}$<br>$l = 11.4 \times 200 = 2280 \text{ cm}$<br><br>$960 \div 30 = 32 \text{ pavers wide}$<br>$2280 \div 30 = 76 \text{ pavers long}$<br><br>$32 \times 76 = 2432 \text{ pavers were used.}$ |
| Specific behaviours                                                                                                                                                                                                              |
| ✓ uses scale correctly<br>✓ indicates appropriate method<br>✓ correct number of pavers                                                                                                                                           |

**Question 11**

**(5 marks)**

A single person living away from home and looking for a job qualifies for a government allowance of \$496.50 per fortnight, provided their fortnightly income is no more than \$150.

In any fortnight that such a person earns more than \$150, the allowance will reduce by 50 cents for each dollar of income they have between \$150 and \$250. If their income is over \$250, the payment will further reduce by 60 cents for each dollar of income over \$250.

Chi qualifies for the allowance. Each week he works two, 5-hour shifts and one, 4-hour shift at a hotel that pays \$20.75 per hour.

(a) Calculate his fortnightly earnings.

**(2 marks)**

| <b>Solution</b>                                 |
|-------------------------------------------------|
| Hours = $(2 \times 5 + 4) \times 2 = 28$        |
| Earnings = $28 \times 20.75 = \$581$            |
| <b>Specific behaviours</b>                      |
| ✓ indicates correct hours worked each fortnight |
| ✓ calculates earnings                           |

(b) Determine his fortnightly government allowance.

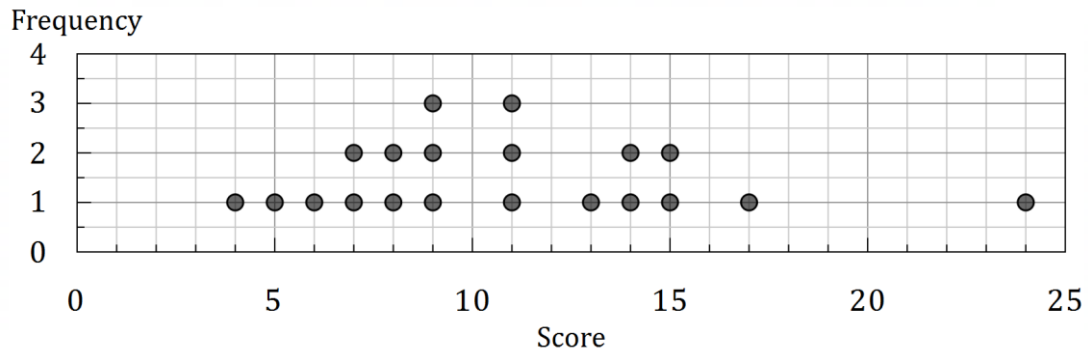
**(3 marks)**

| <b>Solution</b>                                                         |
|-------------------------------------------------------------------------|
| Reduction 1 = $(250 - 150) \times 0.5 = \$50$                           |
| Reduction 2 = $(581 - 250) \times 0.6$<br>= $331 \times 0.6 = \$198.60$ |
| Allowance = $496.50 - 50 - 198.60$<br>= $\$247.90$                      |
| <b>Specific behaviours</b>                                              |
| ✓ indicates first reduction                                             |
| ✓ indicates second reduction                                            |
| ✓ correct allowance                                                     |

## Question 12

(10 marks)

A class of students sat a general knowledge quiz that had 25 multiple choice questions, each with one correct answer. The score of each student in the quiz is shown in the dot plot below.



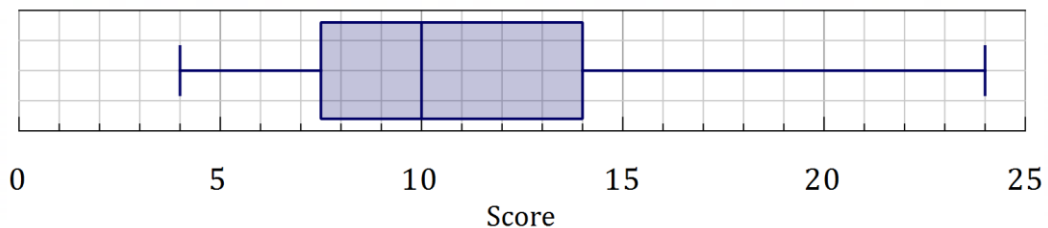
- (a) State the number of students in the class. (1 mark)

| Solution            |
|---------------------|
| 20 students.        |
| Specific behaviours |
| ✓ correct number    |

- (b) Name the statistical variable shown in the dot plot and classify this variable as ordinal, nominal, discrete or continuous. (2 marks)

| Solution                                                       |
|----------------------------------------------------------------|
| Variable is <b>score</b> . Classification is <b>discrete</b> . |
| Specific behaviours                                            |
| ✓ correct variable                                             |
| ✓ correct classification                                       |

- (c) Represent the data from the dot plot as a box plot on the axes below. (4 marks)



| Solution                                                                          |
|-----------------------------------------------------------------------------------|
| Minimum is 4 and maximum is 24.<br>Median is 10.<br>$Q_1$ is 7.5 and $Q_3$ is 14. |
| Specific behaviours                                                               |
| ✓ indicates correct minimum and maximum                                           |
| ✓ indicates correct median                                                        |
| ✓ indicates correct quartiles                                                     |
| ✓ uses values to draw box plot neatly                                             |

- (d) State the interquartile range for this data and use it in a calculation to determine whether the highest score can be classified as an outlier. (3 marks)

| Solution                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $IQR = 14 - 7.5 = 6.5$ $Q_3 + 1.5 \times IQR = 14 + 1.5 \times 6.5 = 14 + 9.75 = 23.75$ <p>Hence any score above 23.75 is an outlier, and so the highest score in the dataset of 24 is an outlier.</p> |
| Specific behaviours                                                                                                                                                                                    |
| <ul style="list-style-type: none"><li>✓ correct IQR</li><li>✓ calculates <math>Q_3 + 1.5 \times IQR</math></li><li>✓ reasons that highest score is an outlier</li></ul>                                |

## Question 13

(8 marks)

The current share price of a resources company is \$6.85.

- (a) A shareholder owns 880 shares in the company, and it has announced a dividend per share of 4.5% of the share price. Determine the dividend the investor will receive.

(2 marks)

| Solution                                             |
|------------------------------------------------------|
| Dividend per share = $0.045 \times 6.85 = \$0.30825$ |
| Dividend = $0.30825 \times 880 = \$271.26$           |
| Specific behaviours                                  |
| ✓ indicates dividend per share                       |
| ✓ correct dividend                                   |

- (b) Determine the total cost for the shareholder to buy another 250 shares in the company at their current price, given that their broker charges a fee of \$40 plus a commission of 1.6% of the value of the shares bought.

(2 marks)

| Solution                                               |
|--------------------------------------------------------|
| Value = $250 \times 6.85 = \$1712.50$                  |
| % commission = $1712.50 \times 0.016 = \$27.40$        |
| Total cost = $1712.50 + 40 + 27.40 = \$1779.90$        |
| Specific behaviours                                    |
| ✓ indicates value of shares                            |
| ✓ indicates percentage commission and calculates total |

- (c) Over the course of a year, the share price of the company increased from \$5.31 to its current price. Determine the percentage increase in the share price.

(2 marks)

| Solution                                          |
|---------------------------------------------------|
| Increase = $6.85 - 5.31 = 1.54$                   |
| % Increase = $1.54 \div 5.31 \times 100 = 29.0\%$ |
| Specific behaviours                               |
| ✓ indicates increase in value                     |
| ✓ correct percentage increase                     |

- (d) The company reported that its latest earnings per share was 41.5 cents. Determine the price to earnings ratio for this company.

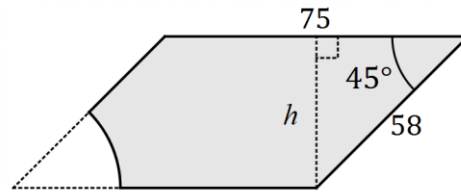
(2 marks)

| Solution                        |
|---------------------------------|
| P/E = $685 \div 41.5 = 16.5$    |
| Specific behaviours             |
| ✓ indicates correct calculation |
| ✓ correct value                 |

**Question 14**

**(7 marks)**

The shape of a metal reinforcing plate is a parallelogram with a sector of a circle removed, as shown in the diagram (not to scale). The acute angle between the sides of 75 cm and 58 cm is  $45^\circ$ , and the radius of the sector removed is 32 cm.



- (a) Show that the perpendicular distance  $h$  between the two long sides is 41 cm. (1 mark)

| Solution                                           |
|----------------------------------------------------|
| $h = 58 \times \sin(45^\circ)$ $= 41.0 \text{ cm}$ |
| Specific behaviours                                |
| ✓ shows correct trig expression                    |

- (b) Determine the area of the sector removed from the parallelogram. (2 marks)

| Solution                                                         |
|------------------------------------------------------------------|
| $A = \frac{45}{360} \times \pi \times 32^2$ $= 402 \text{ cm}^2$ |
| Specific behaviours                                              |
| ✓ indicates correct fraction of circle                           |
| ✓ calculates area                                                |

- (c) Determine the area of the metal plate. (2 marks)

| Solution                                            |
|-----------------------------------------------------|
| Parallelogram area: $A = 75 \times 41 = 3075$       |
| Area of plate: $A = 3075 - 402 = 2673 \text{ cm}^2$ |
| Specific behaviours                                 |
| ✓ area of parallelogram                             |
| ✓ area of plate                                     |

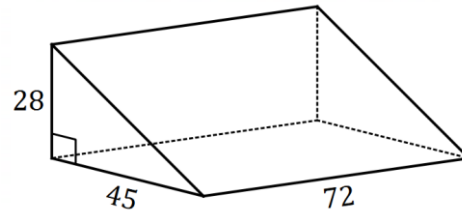
- (d) Every edge of the plate is welded to the main structure. Determine the total length of the five edges of the plate. (2 marks)

| Solution                                                            |
|---------------------------------------------------------------------|
| Curved edge:                                                        |
| $L = \frac{45}{360} \times 2\pi \times 32 = 25.1$                   |
| Total length:                                                       |
| $L = 25 + 2 \times 75 + 2 \times 58 - 2 \times 32 = 227 \text{ cm}$ |
| Specific behaviours                                                 |
| ✓ length of curved edge                                             |
| ✓ calculates total length                                           |

**Question 15****(8 marks)**

A plastic wedge is shown, not to scale.

It takes the form of a prism, with a length of 72 mm and a right triangular cross section measuring 28 mm by 45 mm.



- (a) Calculate the volume of the plastic wedge.

**(2 marks)**

| Solution                                                                                        |
|-------------------------------------------------------------------------------------------------|
| $A = 0.5 \times 45 \times 28 = 630$<br>$V = 630 \times 72 = 45\,360 \text{ mm}^3$               |
| Specific behaviours                                                                             |
| <ul style="list-style-type: none"> <li>✓ area of triangle</li> <li>✓ volume of wedge</li> </ul> |

- (b) Calculate the total surface area of the plastic wedge.

**(4 marks)**

| Solution                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Let <math>x</math> be length of hypotenuse of triangle.</p> $x = \sqrt{28^2 + 45^2} = 53$ <p>Area of rectangles:</p> $A = 72 \times 53 + 72 \times 45 + 72 \times 28 = 9072$ <p>Area of triangles:</p> $A = 2 \times 630 = 1260$ <p>TSA:</p> $A = 9072 + 1260 = 10\,332 \text{ mm}^2$ |
| Specific behaviours                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>✓ hypotenuse of triangle</li> <li>✓ area of rectangles</li> <li>✓ area of triangles</li> <li>✓ correct TSA</li> </ul>                                                                                                                             |

- (c) A scale model of the plastic wedge shown has a length of 24 mm. Determine the total surface area of the scale model.

**(2 marks)**

| Solution                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------|
| <p>Scale factor:</p> $f = 72 \div 24 = 3 \text{ times smaller}$ <p>TSA of model:</p> $A = 10\,332 \div 3^2 = 1148 \text{ mm}^2$ |
| Specific behaviours                                                                                                             |
| <ul style="list-style-type: none"> <li>✓ indicates scale factor</li> <li>✓ correct TSA</li> </ul>                               |

Question 16

(7 marks)

The table below shows the number of tickets sold for seats in the stalls (S), circle (C) and upper circle (U) of a theatre for a show on Tuesday night and Wednesday night.

| Tuesday |     |     | Wednesday |     |     |
|---------|-----|-----|-----------|-----|-----|
| S       | C   | U   | S         | C   | U   |
| 315     | 294 | 198 | 323       | 283 | 206 |

- (a) Represent this information in the  $2 \times 3$  matrix  $A$ , so that  $A_{1,1}$  is the number of stalls tickets sold for Tuesday and  $A_{2,3}$  is the number of upper circle tickets sold for Wednesday.

(2 marks)

| Solution                                                               |
|------------------------------------------------------------------------|
| $A = \begin{bmatrix} 315 & 294 & 198 \\ 323 & 283 & 206 \end{bmatrix}$ |
| Specific behaviours                                                    |
| ✓ $A_{1,1}$ and $A_{2,3}$ correct<br>✓ all coefficients correct        |

Ticket prices to sit in the stalls, circle and upper circle were \$92, \$135 and \$75 respectively.

- (b) Represent the ticket price information in matrix  $B$  so that matrices  $A$  and  $B$  can be multiplied together to create a meaningful result.

(1 mark)

| Solution                                            |
|-----------------------------------------------------|
| $B = \begin{bmatrix} 92 \\ 135 \\ 75 \end{bmatrix}$ |
| Specific behaviours                                 |
| ✓ correct matrix $B$                                |

- (c) State the order in which matrices  $A$  and  $B$  must be multiplied together, and determine the result of this multiplication, matrix  $M$ .

(2 marks)

| Solution                                                                            |
|-------------------------------------------------------------------------------------|
| Order is $A \times B$ .<br>$M = \begin{bmatrix} 83 & 520 \\ 83 & 371 \end{bmatrix}$ |
| Specific behaviours                                                                 |
| ✓ clearly states correct order<br>✓ correct matrix $M$                              |

- (d) State the night on which the value of ticket sales was highest and calculate by how much the sales on this night exceeded the other night.

(2 marks)

| Solution                                                                        |
|---------------------------------------------------------------------------------|
| $83\ 520 - 83\ 371 = 149$<br><br>Sales were highest on Tuesday night, by \$149. |
| Specific behaviours                                                             |
| ✓ correct night<br>✓ correct amount                                             |

## Question 17

(10 marks)

The retirement ages of a group of 80 people is summarised in the following table.

| Age ( $t$ years) | Frequency $f$ |
|------------------|---------------|
| $48 < t \leq 51$ | 1             |
| $51 < t \leq 54$ | 0             |
| $54 < t \leq 57$ | 2             |
| $57 < t \leq 60$ | 5             |
| $60 < t \leq 63$ | 16            |
| $63 < t \leq 66$ | 28            |
| $66 < t \leq 69$ | 23            |
| $69 < t \leq 72$ | 5             |

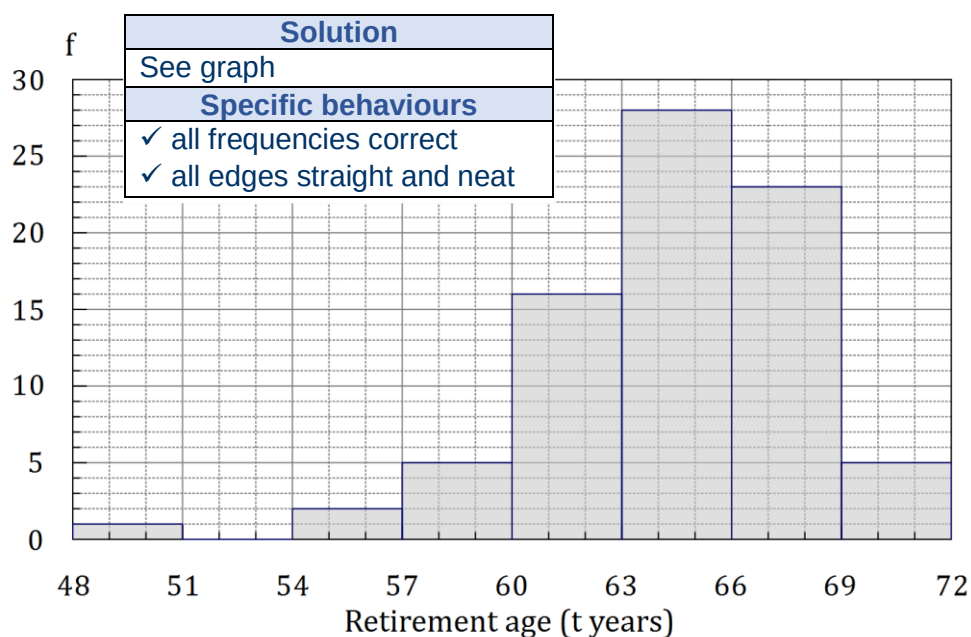
- (a) A student looked at the above data and stated that the oldest person of the group who retired was aged 72. Comment on the validity of this statement. (2 marks)

| Solution                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Statement is not valid. The summary shows that 5 people aged from 69 to 72 retired and it is quite possible that none of these were actually aged 72. |
| Specific behaviours                                                                                                                                   |
| <ul style="list-style-type: none"> <li>✓ states not a valid statement</li> <li>✓ reasonable explanation</li> </ul>                                    |

- (b) Use the mid-point of each class interval to determine the mean and standard deviation of the retirement ages. (2 marks)

| Solution                                                                               |
|----------------------------------------------------------------------------------------|
| $\bar{t} = 64.35 \text{ y}$<br>$\text{sd} = 3.70 \text{ y}$                            |
| Specific behaviours                                                                    |
| <ul style="list-style-type: none"> <li>✓ mean</li> <li>✓ standard deviation</li> </ul> |

- (c) Construct a histogram of the retirement ages on the axes below. (2 marks)



- (d) Describe the distribution of retirement ages in terms of shape and modality. (2 marks)

| Solution                                                                               |
|----------------------------------------------------------------------------------------|
| The shape of the distribution exhibits negative skew.<br>The distribution is unimodal. |
| Specific behaviours                                                                    |
| ✓ negative skew<br>✓ unimodal                                                          |

- (e) If the outlier was removed from the dataset, describe how your answers to part (b) would change. (2 marks)

| Solution                                                               |
|------------------------------------------------------------------------|
| The mean would increase.<br>The standard deviation would decrease.     |
| Specific behaviours                                                    |
| ✓ describes change to mean<br>✓ describes change to standard deviation |

**Question 18****(8 marks)**

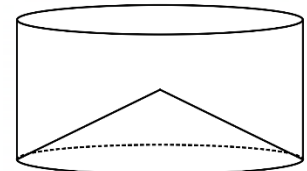
A solid cone has a radius of 9.1 cm and a perpendicular height of 6 cm.

- (a) Determine the slant height of the cone and hence its total surface area.

**(3 marks)**

| Solution                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $s = \sqrt{9.1^2 + 6^2} = 10.9 \text{ cm}$ $A = \pi(9.1)(10.9) + \pi(9.1)^2$ $= 311.6 + 260.2$ $= 572 \text{ cm}^2$                                            |
| Specific behaviours                                                                                                                                            |
| <ul style="list-style-type: none"> <li>✓ calculates slant height</li> <li>✓ calculates curved surface area</li> <li>✓ calculates total surface area</li> </ul> |

The cone is placed inside a hollow cylinder of radius 9.1 cm and height 8.8 cm as shown in the diagram (not to scale), and the remaining space inside the cylinder filled with liquid.



- (b) Determine the volume of liquid that is required, rounding your answer to the nearest  $10 \text{ cm}^3$ .

**(5 marks)**

| Solution                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $V_{\text{CONE}} = \frac{1}{3}\pi(9.1)^2(6)$ $= 520$ $V_{\text{CYL}} = \pi(9.1)^2(8.8)$ $= 2289$ $V_L = 2289 - 520$ $= 1769$ $\approx 1770 \text{ cm}^3$                                                                                                               |
| Specific behaviours                                                                                                                                                                                                                                                    |
| <ul style="list-style-type: none"> <li>✓ substitutes into expressions for cone and cylinder volume</li> <li>✓ evaluates cone volume</li> <li>✓ evaluates cylinder volume</li> <li>✓ evaluates volume of liquid</li> <li>✓ rounds as required, stating units</li> </ul> |

**Question 19**

**(8 marks)**

Mary bought an artwork as an investment at an auction for \$85 000. The auction house charged Mary a handling fee of 0.95% of the sale price.

- (a) Determine the handling fee paid by Mary.

(1 mark)

| Solution                                     |
|----------------------------------------------|
| $\frac{0.95}{100} \times 85\,000 = \$807.50$ |
| Specific behaviours                          |
| ✓ correct amount                             |

One year later, Mary sold the artwork through a gallery for \$129 200. The gallery charged Mary a commission of 32% of the sale price.

- (b) Determine the net profit made by Mary, after she had paid the handling fee and sales commission.

(3 marks)

| Solution                                              |
|-------------------------------------------------------|
| $\frac{32}{100} \times 129\,200 = \$41\,344$          |
| Profit = $129\,200 - 85\,000 = \$44\,200$             |
| Net Profit = $44\,200 - 807.50 - 41\,344 = \$2048.50$ |
| Specific behaviours                                   |
| ✓ calculates commission                               |
| ✓ calculates profit                                   |
| ✓ calculates net profit                               |

At the time Mary bought the artwork, she could have placed the money she paid for it in a 12 month term deposit that offered an interest rate of 3.3% compounded monthly.

- (c) Determine whether the term deposit would have been a better investment for Mary.

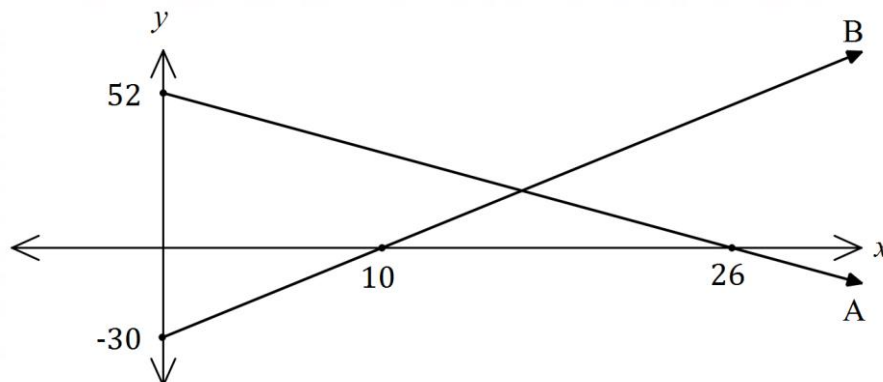
(4 marks)

| Solution                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F = 85\,000 \left(1 + \frac{3.3}{12 \times 100}\right)^{12}$ $= \$87\,847.82$ $I = 87\,847.82 - 85\,000 = \$2847.82$ <p>Hence term deposit would have been a better investment as interest was nearly \$800 more than net profit Mary gained on artwork.</p> |
| Specific behaviours                                                                                                                                                                                                                                           |
| ✓ shows correct use of compound interest formula<br>✓ calculates future value of deposit<br>✓ calculates interest, to nearest cent<br>✓ appropriate response to question using calculated values                                                              |

## Question 20

(8 marks)

The temperatures of component A and component B in a heat exchanger vary in a linear fashion as shown in the graph below, where  $y$  is the temperature in  $^{\circ}\text{C}$ , after  $x$  minutes.



- (a) Determine a rule for the temperature of component A in terms of  $x$ .

(3 marks)

| Solution                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| $m = -52 \div 26 = -2$<br>$c = 52$<br>$y = -2x + 52$                                                                                        |
| Specific behaviours                                                                                                                         |
| <ul style="list-style-type: none"> <li>✓ correct gradient</li> <li>✓ correct intercept</li> <li>✓ writes rule for <math>y</math></li> </ul> |

- (b) Explaining your method, determine when both components are at the same temperature, and state what this temperature is.

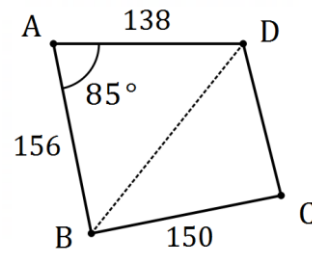
(5 marks)

| Solution                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Require equation for component B:</p> $m = 30 \div 10 = 3$<br>$c = -30$<br>$y = 3x - 30$                                                                                                             |
| <p>Now use calculator to graph both lines and find point of intersection at <math>(16.4, 19.2)</math>.</p>                                                                                              |
| <p>Hence components reach the same temperature of <math>19.2^{\circ}\text{C}</math> after 16.4 minutes.</p>                                                                                             |
| Specific behaviours                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>✓ correct gradient for line B</li> <li>✓ writes rule for B</li> <li>✓ states time</li> <li>✓ states temperature</li> <li>✓ explains method throughout</li> </ul> |

**Question 21**

**(8 marks)**

Field  $ABCD$  is a quadrilateral with sides  $AB, BC$  and  $AD$  having lengths 156, 150 and 138 m respectively, as shown in the diagram (not to scale). The size of angle  $BAD$  is  $85^\circ$ .



In this question, you must justify your answers by showing use of trigonometry.

- (a) Determine the length of the diagonal  $BD$ .

**(2 marks)**

| Solution                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|
| $BD^2 = 128^2 + 156^2 - 2(138)(156) \cos 85^\circ$ $BD = 199 \text{ m}$                                     |
| Specific behaviours                                                                                         |
| <ul style="list-style-type: none"> <li>✓ correct use of cosine rule</li> <li>✓ calculates length</li> </ul> |

- (b) Determine the size of angle  $ABD$ .

**(2 marks)**

| Solution                                                                                                 |
|----------------------------------------------------------------------------------------------------------|
| $\frac{\sin \angle ABD}{138} = \frac{\sin 85^\circ}{199}$ $\angle ABD = 43.7^\circ$                      |
| Specific behaviours                                                                                      |
| <ul style="list-style-type: none"> <li>✓ correct use of sine rule</li> <li>✓ calculates angle</li> </ul> |

It is also known that the size of angle  $ABC$  is  $76^\circ$ .

- (c) Calculate the size of angle  $DBC$ .

**(1 mark)**

| Solution                                                             |
|----------------------------------------------------------------------|
| $\angle DBC = 76 - 43.7 = 32.3^\circ$                                |
| Specific behaviours                                                  |
| <ul style="list-style-type: none"> <li>✓ calculates angle</li> </ul> |

- (d) Determine the area of the field, rounding your answer to the nearest 100 square metres.

**(3 marks)**

| Solution                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A_{ABD} = \frac{1}{2}(138)(156) \sin 85^\circ = 10\,723$ $A_{CBD} = \frac{1}{2}(199)(150) \sin 32.3^\circ = 7983$ $\text{Area} = 10\,723 + 7983 = 18\,706$ $\approx 18\,700 \text{ m}^2$ |
| Specific behaviours                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>✓ area of one triangle, showing trigonometry</li> <li>✓ area of second triangle</li> <li>✓ calculates total area</li> </ul>                        |

Supplementary page

Question number: \_\_\_\_\_

Supplementary page

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